Ruukkilabra
UPPER SECONDARY PHYSICS

FY2

Physics, environment and society

KIVIRUUKIN LUKIO

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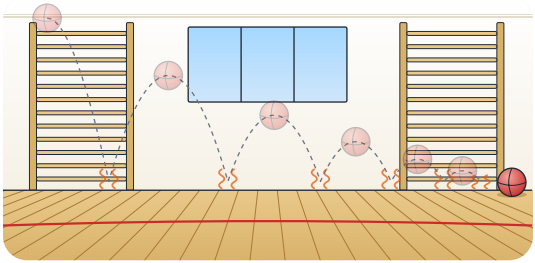
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FY2 · CHAPTER 1

Forms of energy and conservation of energy

Why does a bouncing ball eventually come to rest on the floor?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy2-1.html`



IN THIS LESSON, YOU WILL LEARN TO

- name the forms of energy and recognise which form the energy is in.
- calculate an unknown energy using the law of conservation of energy $E_{\text{initial}} = E_{\text{final}}$.
- explain why using up energy means changing it, not losing it.

PRIOR KNOWLEDGE

- The prefix kilo means a thousand, so $1\,\text{kJ} = 1000\,\text{J}$
  chapter *Quantities, units and the SI system*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 1.

Think about it

A bouncing ball is dropped on the floor. The first bounce rises almost as high as the height from which the ball was dropped, the next bounce is lower and the next one lower still. In the end the ball lies still on the floor. At the start the ball had a lot of energy high up, but at the end it is neither high up nor moving.

> **Where did the energy of the ball go?**

Theory

Key idea

is a quantity used to describe what happens in a system when something moves, warms up, gives off light or changes its form. The symbol for energy is E and its unit is the , J. The joule is a small unit, so energies are often given in kilojoules too: the prefix kilo means a thousand, so $1\,\mathrm{kJ} = 1000\,\mathrm{J}$.

Energy comes in many forms, and each form is called a . The form of energy tells you where the energy is stored or how it shows itself.



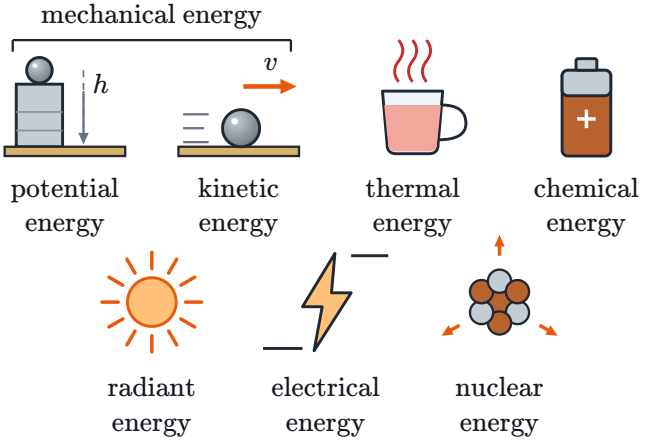
The site of a hydroelectric power plant seen from above. In a hydroelectric plant the potential energy of water turns into kinetic energy in the turbine and into electrical energy in the generator. Photo: SHADY, Pexels License

Seven forms of energy

- E_p is energy that depends on the position of an object, here gravitational potential energy. The higher an object is and the greater its mass, the more potential energy it has. A ball at the edge of a shelf can pick up speed as it falls.
- E_k is the energy of a moving object. The faster an object moves and the greater its mass, the more kinetic energy it has. An object at rest has none.
- **Thermal energy** E_thermal is energy linked to the motion of the particles of a substance and to the interactions between them. When an object warms up, its particles move more and its thermal energy increases. Thermal energy also depends on the amount of substance, so it is not the same thing as temperature. The word , on the other hand, means in physics energy that is *transferred* because of a temperature difference: the net transfer of energy is from the warmer object to the colder one. An object has thermal energy, and energy is transferred from one object to another as heat.
- is energy stored in the chemical bonds of substances. It is found in food, fuels and batteries, and it is released in, for example, combustion and metabolism.

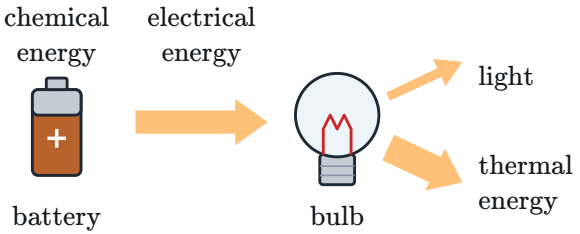
- is energy carried by radiation, for example sunlight. It also travels through a vacuum.
- is energy transferred by an electric current, for example in the electricity grid.
- is energy stored in atomic nuclei. It is released in nuclear reactions, which take place in a nuclear power plant and in the Sun.

Potential energy and kinetic energy together make up , $E_{\text{mech}} = E_{\text{p}} + E_{\text{k}}$.



Energy changes form

Energy does not stay in the same form. In a torch (flashlight), the chemical energy of the battery changes into electrical energy, and the bulb changes the electrical energy into light and thermal energy. In an incandescent bulb, most of the energy changes into thermal energy and only a small part into light.



There are similar changes everywhere. A plant changes the radiant energy of the Sun into chemical energy. A muscle changes chemical energy into kinetic energy and thermal energy. In a nuclear power plant, nuclear energy finally changes into electrical energy.

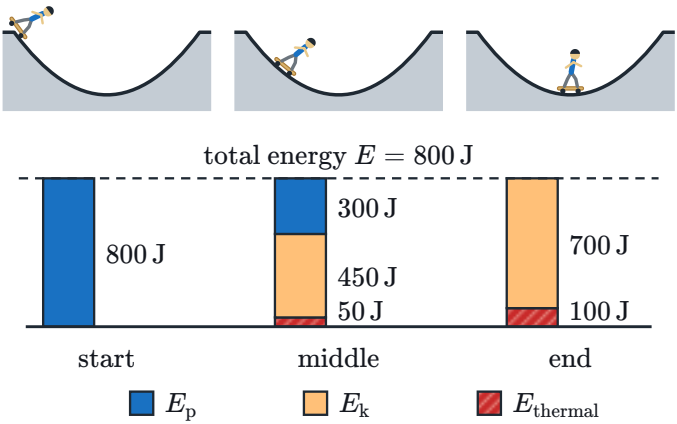
Law of conservation of energy

In all these changes the total amount of energy stays the same: energy is neither created nor destroyed, but changes form or moves from one place to another. This is a universal principle of physics, the . We look at an **isolated system**, which receives no energy from outside and loses none to the outside. Then the total energy of the system is the same at the start and at the end:

$$E_{\text{initial}} = E_{\text{final}}$$

where E_{initial} is the sum of the energies of all forms of energy in the system at the start and E_{final} is their sum at the end. Both are given in joules (J).

The sum includes all forms of energy, thermal energy too. When a skateboarder slides from the edge of a half-pipe to the bottom and friction acts, the mechanical energy $E_{\text{p}} + E_{\text{k}}$ decreases, but the thermal energy E_{thermal} increases by the same amount. The sum $E_{\text{p}} + E_{\text{k}} + E_{\text{thermal}}$ does not change. So friction does not create energy: it changes mechanical energy into thermal energy, and the wheels, the ramp and the air warm up a little.



Using up energy is changing energy

In everyday language energy is "used up", but in physics it does not run out. When a device uses energy, it changes energy from one form to another. Most often the largest part changes into thermal energy, which is transferred as heat into the surroundings and is hard to use any more. In short, people often say that energy "changes into heat". This is why it is worth saving energy even though it cannot be destroyed.

Common misconception

Energy does not disappear or get used up. When a ball stops or a device uses energy, the energy has changed into another form, usually thermal energy that has been transferred as heat into the surroundings. Another mistake is to calculate only the mechanical energy. In motion with friction $E_\mathrm{p} + E_\mathrm{k}$ decreases, but the total energy stays the same when thermal energy is included. Friction does not create energy but changes it from one form to another.

Summary

- The symbol for energy is E and its unit is the joule J; $1\,\mathrm{kJ} = 1000\,\mathrm{J}$.
- The forms of energy are potential energy, kinetic energy (together mechanical energy), thermal energy, chemical energy, radiant energy, electrical energy and nuclear energy.
- Energy changes form and moves but is neither created nor destroyed: in an isolated system $E_\mathrm{initial} = E_\mathrm{final}$.
- Friction does not create energy but changes mechanical energy into thermal energy: $E_\mathrm{p} + E_\mathrm{k}$ decreases, but the total energy does not change.
- An object has thermal energy; heat is energy that is transferred because of a temperature difference, net from the warmer object to the colder one.
- Using up energy means changing it into a form that is hard to use, usually thermal energy.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Skateboarder in a half-pipe

A skateboarder starts from rest at the edge of a half-pipe. At the edge her potential energy is $840\,\mathrm{J}$, when the zero level is the bottom of the half-pipe. At the bottom her kinetic energy is $0.72\,\mathrm{kJ}$. The skateboarder, the half-pipe and the surroundings form an isolated system. How much energy has friction changed into thermal energy during the slide?

STEP 1/7

Given and find. At the start the potential energy is $E_{\mathrm{p}} = 840\,\mathrm{J}$ and the kinetic energy is zero, because the skateboarder is at rest. At the bottom the potential energy is zero and the kinetic energy is $E_{\mathrm{k}} = 0.72\,\mathrm{kJ}$. We want the amount of mechanical energy E_{thermal} that friction changes into thermal energy during the slide.

STEP 2/7

Principle. The system is isolated, so energy is conserved: $E_{\mathrm{initial}} = E_{\mathrm{final}}$. At the start all the energy is potential energy. At the bottom it is divided between kinetic energy and thermal energy, into which friction has changed part of it.

STEP 3/7

Rearranged formula. At the start there is only E_{p}, and at the end there are E_{k} and E_{thermal}:

$$\begin{aligned} E_{\mathrm{initial}} &= E_{\mathrm{final}} \\ E_{\mathrm{p}} &= E_{\mathrm{k}} + E_{\mathrm{thermal}} \\ E_{\mathrm{thermal}} &= E_{\mathrm{p}} - E_{\mathrm{k}} \end{aligned}$$

STEP 4/7

Unit conversions. The kinetic energy is given in kilojoules and the potential energy in joules. The prefix kilo means a thousand, so it is put in place of the unit:

$$\begin{aligned} E_{\mathrm{k}} &= 0.72\,\mathrm{kJ} \\ &= 0.72 \cdot 1000\,\mathrm{J} \\ &= 720\,\mathrm{J} \end{aligned}$$

STEP 5/7

Substitution.

$$\begin{aligned} E_{\mathrm{thermal}} &= E_{\mathrm{p}} - E_{\mathrm{k}} \\ &= 840\,\mathrm{J} - 720\,\mathrm{J} \\ &= 120\,\mathrm{J} \end{aligned}$$

STEP 6/7

Result. Friction changed $E_{\text{thermal}} = 120\,\text{J}$ into thermal energy. The result is rounded to tens of joules, because the kinetic energy $0.72\,\text{kJ}$ is given to the nearest ten joules and this limits the precision of the difference.

STEP 7/7

Reasonableness. The thermal energy is positive and smaller than the initial energy $840\,\text{J}$, as it should be with friction. The energy is accounted for:

$$\begin{aligned} 720\,\text{J} + 120\,\text{J} &= 840\,\text{J} \\ &= E_{\text{initial}} \end{aligned}$$

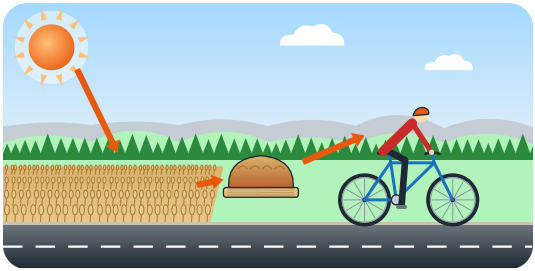
Less than a fifth of the initial energy changed into thermal energy, which is a believable amount on a smooth half-pipe.

FY2 · CHAPTER 2

Energy transformations and the joule

Why does the motion of a bicycle ultimately come from the Sun, and how can the amount of energy be given as a single number?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy2-2.html`



IN THIS LESSON, YOU WILL LEARN TO

- explain an energy chain and what the energy finally turns into.
- give energy in joules and with the prefixes kJ, MJ, GJ and TJ.
- compare everyday and societal energies by order of magnitude.

PRIOR KNOWLEDGE

- Energy appears in different forms and is conserved
  chapter *Forms of energy and conservation of energy*
- The prefixes kilo, mega, giga and tera are powers of ten
  chapter *Quantities, units and the SI system*

- Powers of ten and orders of magnitude
  chapter *Physics as a science and the scales of the universe*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 4.

Think about it

In the morning you eat oat porridge and set off to school by bike. The pedals turn, your speed stays up and the tyres warm up. When you arrive, you brake and stop. The oats grew in a field through the summer, and the Sun shone on them the whole time.

> **Where does the energy of the bicycle's motion come from, and where does it finally go when you stop?**

Theory

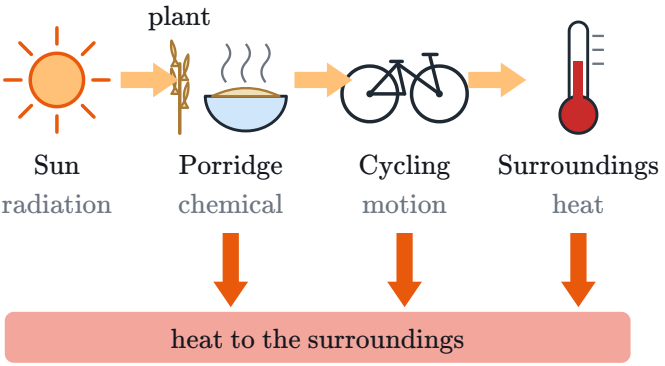
Key idea

Energy changes from one form to another

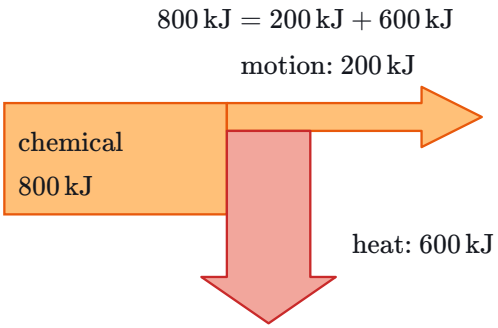
Energy is a quantity that describes what can happen in a system. It appears in different forms, such as mechanical, chemical, electrical, thermal energy, radiation and nuclear energy (chapter *Forms of energy and conservation of energy*). In everyday life, energy seldom stays in one form: it changes from one form to another in a chain of steps.

The cyclist's chain begins with the Sun. Radiation energy is stored in a plant as chemical energy, and it ends up in oats and in morning porridge. The muscles turn part of the chemical energy of the porridge into the motion of the bicycle, and

when you stop, the kinetic energy finally turns into thermal energy of the tyres, the brakes and the air.



At every step, part of the energy turns into thermal energy, which is transferred as heat to the surroundings and spreads out. Even so, no energy disappears: as stated in chapter *Forms of energy and conservation of energy*, there is the same amount of energy at the start and at the end of the chain. In the example in the figure, the $800\ \mathrm{kJ}$ of the porridge is divided into motion and thermal energy, and the parts add up to the same amount as at the start. Thermal energy spreads so widely that it is no longer easy to put to use, and this is why energy seems to "run out".



the width of the arrow shows the amount of energy

Joule

The symbol of energy is E and its unit is the joule, whose symbol is J. One joule is a small amount of energy: it is roughly what is needed to lift an apple of about $0.1\ \mathrm{kg}$ by one metre. In this chapter you learn the size of the joule through examples.



Launch of the Artemis I rocket in Florida in November 2022. The chemical energy of the fuel turns into the rocket's kinetic energy, heat, light and sound.

Photo: NASA/Joel Kowsky, public domain

Unit of energy and its multiples

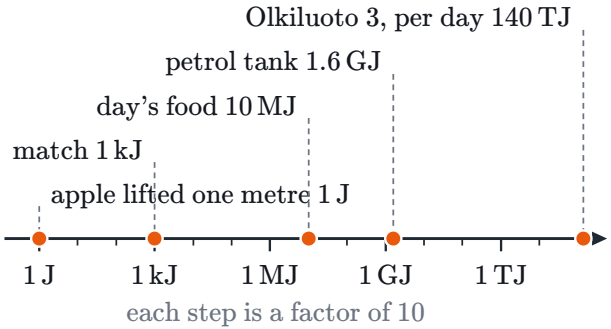
Everyday energies are large numbers when given in joules, so we use prefixes (chapter *Quantities, units and the SI system*): kilo (k), mega (M), giga (G) and tera (T). Each of them is a thousand times the previous one:

$$1\,\mathrm{kJ} = 1000\,\mathrm{J}$$
$$1\,\mathrm{MJ} = 1000\,\mathrm{kJ}$$
$$1\,\mathrm{GJ} = 1000\,\mathrm{MJ}$$
$$1\,\mathrm{TJ} = 1000\,\mathrm{GJ}$$

where J is the joule, kJ the kilojoule, MJ the megajoule, GJ the gigajoule and TJ the terajoule. So a megajoule contains $1000 \cdot 1000\,\mathrm{J} = 1\,000\,000\,\mathrm{J}$.

A conversion is done by replacing the prefix with its value: $2.5\,\mathrm{kJ} = 2.5 \cdot 1000\,\mathrm{J} = 2500\,\mathrm{J}$. In the other direction, the number is written using the prefix: $3000\,\mathrm{J} = 3 \cdot 1000\,\mathrm{J} = 3\,\mathrm{kJ}$. Energies are added together only in the same unit, so they are first converted to the same unit.

The orders of magnitude of everyday energies are collected in the figure, where each step is a factor of ten (chapter *Physics as a science and the scales of the universe*). When the orders of magnitude of two energies are very different, one division is enough for the comparison: the ratio of the $32\ \mathrm{MJ}$ of a litre of petrol to the $10\ \mathrm{MJ}$ of a day's food is $32\ \mathrm{MJ} : 10\ \mathrm{MJ} \approx 3$.



The amount of energy does not yet tell how fast the energy is used. That is described in the next chapter, *Power*.

Common misconception

Many people think that energy runs out when the speed slows down or the battery goes flat. Energy does not run out and does not disappear. It changes form, and a large part ends up as heat in the surroundings, from where it can no longer easily be collected back for use.

Another mistake is to add energies in different units, for example $1.5\ \mathrm{MJ} + 300\ \mathrm{kJ} = 301.5$. A prefix is a multiplier, so the energies are first converted to the same unit.

Summary

- Energy changes from one form to another in a chain, for example radiation, chemical energy, kinetic energy and thermal energy.
- Energy is conserved (chapter *Forms of energy and conservation of energy*). At every step part of it turns into thermal energy, which is transferred as heat to the surroundings.
- The unit of energy is the joule (J). Prefixes: $1\ \mathrm{kJ} = 1000\ \mathrm{J}$, $1\ \mathrm{MJ} = 1000\ \mathrm{kJ}$, $1\ \mathrm{GJ} = 1000\ \mathrm{MJ}$ and $1\ \mathrm{TJ} = 1000\ \mathrm{GJ}$.
- A prefix is replaced by its value: $2.5\ \mathrm{kJ} = 2.5 \cdot 1000\ \mathrm{J}$. Energies are added in the same unit.

- Everyday energies range from joules to terajoules, and they are compared using powers of ten.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Banana energy on a bike

A cyclist eats a banana whose chemical energy is $0.45\ \mathrm{MJ}$. The muscles turn $110\ \mathrm{kJ}$ of it into the motion of cycling, and the rest turns into thermal energy. How much energy turns into thermal energy in the muscles and the body? Give the answer in kilojoules.

STEP 1/7

Given and find. The chemical energy at the start is $E_{\text{start}} = 0.45\ \mathrm{MJ}$ and the energy that turned into motion is $E_{\text{motion}} = 110\ \mathrm{kJ}$. The thermal energy E_{heat} is to be found in kilojoules.

STEP 2/7

Principle. Energy is conserved: the chemical energy of the banana does not disappear but turns entirely into motion and thermal energy. So the energy at the start is equal to the sum of the energies that turned into motion and thermal energy.

STEP 3/7

Rearranged formula.

$$E_{\text{start}} = E_{\text{motion}} + E_{\text{heat}}$$
$$E_{\text{heat}} = E_{\text{start}} - E_{\text{motion}}$$

STEP 4/7

Unit conversions. The energies are calculated in the same unit. Replace the prefix mega with its value:

$$\begin{aligned} 0.45\,\mathrm{MJ} &= 0.45 \cdot 1000\,\mathrm{kJ} \\ &= 450\,\mathrm{kJ} \end{aligned}$$

STEP 5/7

Substitution.

$$\begin{aligned} E_{\text{heat}} &= E_{\text{start}} - E_{\text{motion}} \\ &= 450\,\mathrm{kJ} - 110\,\mathrm{kJ} \\ &= 340\,\mathrm{kJ} \end{aligned}$$

STEP 6/7

Result. The thermal energy produced is $340\,\mathrm{kJ}$. The result is rounded to two significant figures, because the initial energy $0.45\,\mathrm{MJ}$ is given with two significant figures. The $110\,\mathrm{kJ}$ given as a whole number is taken as exact, so the precision is limited by the initial energy.

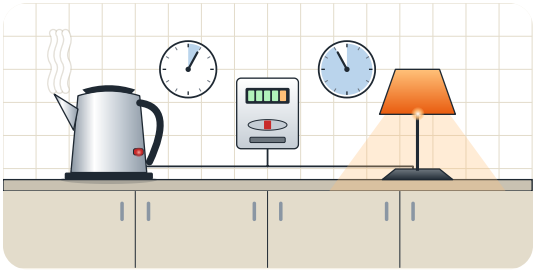
STEP 7/7

Reasonableness. More energy turns into thermal energy than into motion, as is usual in human muscles: the body warms up when you cycle. The sum checks out: $110\,\mathrm{kJ} + 340\,\mathrm{kJ} = 450\,\mathrm{kJ}$, which is the same as at the start.

Power: $P = \Delta E / \Delta t$

Why can you not work out an electricity bill from the power of a device alone?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy2-3.html`



IN THIS LESSON, YOU WILL LEARN TO

- calculate power, energy or time with the formula $P = E : t$.
- explain why an electricity bill depends on both power and time.
- calculate electrical energy in kilowatt-hours and its cost.

PRIOR KNOWLEDGE

- Energy is measured in joules, and it is transformed from one form to another
  chapter *Energy transformations and the joule*

- Energy is neither created nor destroyed
  chapter *Forms of energy and conservation of energy*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 5.

Think about it

In the kitchen a kettle hums and the water heats up in a couple of minutes. At the same time a small lamp is on in the hallway, and nobody needs its light. The reading of the electricity meter rises because of both, but it rises clearly faster because of the kettle. Still, the lamp is often on for hours, while the kettle is on only for a moment.

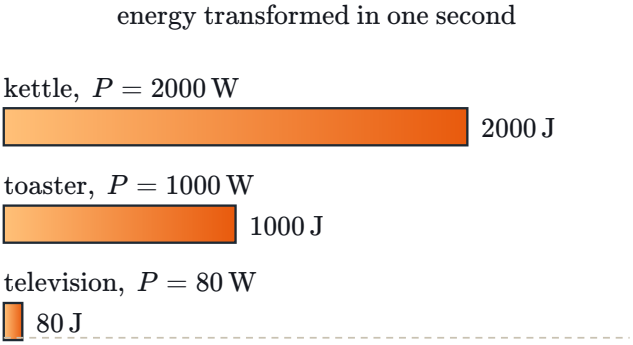
> **Which uses more electrical energy, a device that works fast or a device that is on for a long time, and how can you compare them?**

Theory

Key idea

When a device transforms energy, two things matter: how much energy is transformed and how fast. A kettle transforms electrical energy into heat very quickly, so the water heats up in minutes. A small lamp transforms electrical energy into light and heat slowly.

How fast energy is transformed or transferred is called . Power is energy per unit of time: if a device transforms $2000\,\mathrm{J}$ every second, its power is $2000\,\mathrm{J/s} = 2000\,\mathrm{W}$.



Power does not tell you how much energy a device uses in total. For that you need power together with time: a low power for a long time can transform the same energy as a high power for a short moment. This is why an electricity bill depends on both the power of the device and how long it is used.



Offshore wind turbines at sunset. A turbine's power tells how fast it produces electrical energy. The energy produced in a day also depends on how long and how strongly the wind blows.

Photo: Low Angle, Unsplash License

Power

Power is the quotient of the energy transformed or transferred and the time it takes:

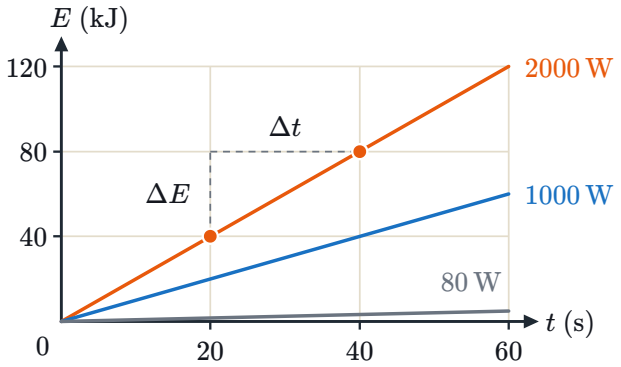
$$P = \frac{\Delta E}{\Delta t}$$

where P is power (W), ΔE is the amount of energy transferred or transformed during the time interval considered (J), and Δt is the length of that interval (s). In words: power is energy divided by time. If energy is transformed unevenly, the formula gives the average power over the time interval Δt. When a device runs steadily from the moment it is switched on, ΔE is the whole transformed energy E and Δt is the whole time t, so $P = E : t$. Energy and time can be solved from the formula:

$$E = P\,t$$
$$t = \frac{E}{P}$$

The unit of power is the watt: $1\ \mathrm{W} = 1\ \mathrm{J/s}$. Prefixes work as with other quantities (chapter *Quantities, units and the SI system*): $1\ \mathrm{kW} = 1000\ \mathrm{W}$ and $1\ \mathrm{MW} = 10^6\ \mathrm{W}$. The power of power plants is measured in megawatts: the electrical power of the Olkiluoto 3 nuclear power unit is about $1600\ \mathrm{MW}$ (TVO), as much as about $800\,000$ kettles.

If a device transforms energy steadily, energy as a function of time is a straight line starting from the origin. The slope of the line is the power: the greater the power, the more steeply the energy grows. In the figure, on the kettle's line the energy grows by $40\ \mathrm{kJ}$ in a time interval of $20\ \mathrm{s}$. Substituting $1\ \mathrm{kJ} = 1000\ \mathrm{J}$: $40\ \mathrm{kJ} = 40 \cdot 1000\ \mathrm{J} = 40\,000\ \mathrm{J}$, so $P = 40\,000\ \mathrm{J} : 20\ \mathrm{s} = 2000\ \mathrm{W}$.



Kilowatt-hour

Electrical energy is billed in kilowatt-hours (kWh). One kilowatt-hour is the energy that a device with a power of $1\,\mathrm{kW}$ transforms in one hour. In joules, an hour is $3600\,\mathrm{s}$ and a kilowatt is $1000\,\mathrm{W}$, so

$$\begin{aligned} 1\,\mathrm{kWh} &= 1000\,\mathrm{W} \cdot 3600\,\mathrm{s} \\ &= 3\,600\,000\,\mathrm{J} \end{aligned}$$

A kilowatt-hour is therefore a unit of energy, not of power. The formula $E = P\,t$ gives kilowatt-hours directly when power is in kilowatts and time in hours: $\mathrm{kW} \cdot \mathrm{h} = \mathrm{kWh}$.

Typical powers and the electricity bill

The table shows typical powers of household appliances, rounded from the typical values given by manufacturers. The power of a real appliance is on its rating plate. The power of a refrigerator is the power while its compressor runs, and the compressor does not run all the time.

Appliance	Power
LED lamp	$8\,\mathrm{W}$
laptop	$50\,\mathrm{W}$
television	$80\,\mathrm{W}$
refrigerator	$100\,\mathrm{W}$
hair dryer	$1800\,\mathrm{W}$
kettle	$2000\,\mathrm{W}$
electric sauna heater	$8\,\mathrm{kW}$

Many appliances use energy even when they seem to be switched off. The standby power is typically from less than a watt to a few watts per appliance. This is small, but standby lasts all $8760\,\mathrm{h}$ of the year.

An electricity bill is the energy consumed times the unit price. In this chapter we use the round example price $20\,\mathrm{c/kWh}$, that is $0.20\,\mathrm{€/kWh}$. This is the order of magnitude of the total price of household electricity in 2025 (Statistics Finland 2025: about $18\,\mathrm{c/kWh}$ as a total price, when energy, transmission and taxes are added together); the example price is rounded up. The real price varies with the contract and the time of year.

Common misconception

Many people think that a high power means high energy consumption. Power only tells the rate: the energy consumed is power times time. A hair dryer ($1800\,\mathrm{W}$) runs for five minutes and consumes $0.15\,\mathrm{kWh}$, but a $10\,\mathrm{W}$ lamp that is on for a day consumes $0.24\,\mathrm{kWh}$. Another mistake is to mix up the units: the watt is the unit of power and the kilowatt-hour the unit of energy, so "the power is $2\,\mathrm{kWh}$" is wrong: the correct statement is that the power is $2\,\mathrm{kW}$.

Summary

- Power tells how fast energy is transformed or transferred: $P = \frac{\Delta E}{\Delta t}$. The unit is the watt, $1\,\mathrm{W} = 1\,\mathrm{J/s}$.
- At steady power $E = P\,t$ and $t = E : P$. In an energy–time graph the slope of the line is the power.
- $1\,\mathrm{kWh} = 1000\,\mathrm{W} \cdot 3600\,\mathrm{s} = 3.6\,\mathrm{MJ}$. The kilowatt-hour is a unit of energy, and an electricity bill is the energy in kWh times the unit price.
- A high power for a short time and a low power for a long time can use the same amount of energy; standby consumes energy all the time.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Electricity bill of a TV

The power of a television is $80\ \mathrm{W}$, and it is on for $4.0\ \mathrm{h}$ a day for $30\ \mathrm{days}$. How much does watching television cost in a month when the price of electricity is $20\ \mathrm{c/kWh}$?

STEP 1/7

Given and find. Power $P = 80\ \mathrm{W}$, usage time $4.0\ \mathrm{h}$ a day for $30\ \mathrm{days}$ and price $20\ \mathrm{c/kWh}$. We want the cost of electricity for the month.

STEP 2/7

Principle. The television transforms energy at a steady power, so the energy consumed is power times time. The cost is found by multiplying the energy in kilowatt-hours by the unit price.

STEP 3/7

Rearranged formula. The energy is $E = P\,t$ and the cost is the energy times the unit price:

$$E = P\,t$$
$$\text{cost} = E \cdot \text{unit price}$$

STEP 4/7

Unit conversions. For kilowatt-hours the power must be in kilowatts. Substitute $1\ \mathrm{W} = 10^{-3}\ \mathrm{kW}$:

$$\begin{aligned} 80\ \mathrm{W} &= 80 \cdot 10^{-3}\ \mathrm{kW} \\ &= 0.080\ \mathrm{kW} \end{aligned}$$

The total time is $4.0\ \mathrm{h} \cdot 30 = 120\ \mathrm{h}$. The cost comes out in cents, and we convert to euros by substituting $1\ \mathrm{c} = 0.01\ €$.

STEP 5/7

Substitution.

$$\begin{aligned} E &= P\,t \\ &= 0.080\,\mathrm{kW} \cdot 120\,\mathrm{h} \\ &= 9.6\,\mathrm{kWh} \\ \text{cost} &= E \cdot \text{unit price} \\ &= 9.6\,\mathrm{kWh} \cdot 20\,\mathrm{c/kWh} \\ &= 192\,\mathrm{c} \\ &= 192 \cdot 0.01\,€ \\ &= 1.92\,€ \end{aligned}$$

STEP 6/7

Result. Watching television costs about $1.9\,€$ a month. The result is given with two significant figures because the power and the daily usage time are given with two figures.

STEP 7/7

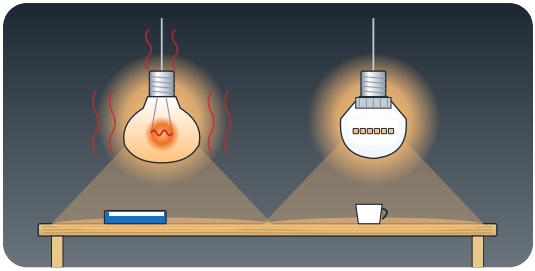
Reasonableness. One kilowatt-hour covers an $80\ \mathrm{W}$ television for about $12\,\mathrm{h}$, and in a month we watch for $120\,\mathrm{h}$, which is about ten kilowatt-hours. That costs about ten times $20\,\mathrm{c}$, which is about two euros.

FY2 · CHAPTER 4

Efficiency

Why does an incandescent lamp get so hot that it can burn your finger, even though its job is to give light, and why does an LED lamp give just as much light with much less electricity?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy2-4.html`



IN THIS LESSON, YOU WILL LEARN TO

- calculate the efficiency of a device as a ratio of energies or powers.
- explain why efficiency is at most $100\ \%$ and why in real devices it is often below that.
- read an energy flow diagram and work out the shares of useful energy and losses from it.

PRIOR KNOWLEDGE

- Energy is neither created nor destroyed. It changes form, and part of it often becomes heat
  chapter *Forms of energy and conservation of energy*
- The unit of energy is the joule, and when energy is transformed, part of it ends up as heat
  chapter *Energy transformations and the joule*
- Power tells how fast energy is transferred, the unit is the watt, and at constant power energy is power multiplied by time
  chapter *Power:* $P = \Delta E / \Delta t$

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 4.

Think about it

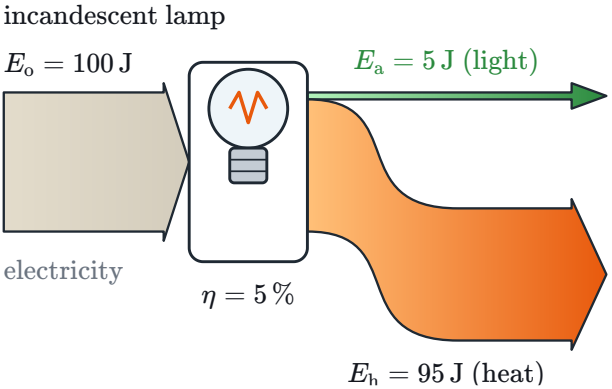
Two lamps are lit in a room, an incandescent lamp and an LED lamp, and they give out the same amount of visible light. The incandescent lamp takes a power of 60 W from the electricity grid, the LED lamp only 10 W. When you touch the lamps after they have been on for a moment, the incandescent lamp is burning hot and the LED lamp is only lukewarm.

> **Where does the electrical energy taken by the incandescent lamp go, and how can you measure how**

Theory

Key idea

Energy does not disappear, but not all the energy taken by a device becomes the form of energy that you want from the device. The job of an incandescent lamp is to give light, so its useful energy is light. However, most of the electrical energy that the lamp takes becomes heat, which warms the lamp and its surroundings but does not give light. This part is called losses. Losses are not lost energy but energy that has changed into a useless form, usually heat.



The flow of energy through a device can be shown with an energy flow diagram, also called a Sankey diagram. In the diagram the thickness of an arrow is proportional to energy: the incoming arrow divides into a thin arrow of useful energy and a thick arrow of losses. When the lamp takes electrical energy of $100\,\mathrm{J}$, only $5\,\mathrm{J}$ becomes light and the remaining $95\,\mathrm{J}$ is heat. The thicknesses of the outgoing arrows add up to the thickness of the incoming arrow, because energy is conserved.

Losses depend on the purpose of the device

In a kettle, heat is useful when it warms the water, and a loss when it escapes through the sides of the kettle into the room. The heat of an incandescent lamp, on the other hand, is counted as a loss, even if it warms a room in winter, because the benefit of the lamp has been defined as light. So what is useful depends on the

purpose of the device, and the value of the efficiency depends on what is defined as useful and what is counted as part of the device.

Efficiency tells the share of the useful part

Efficiency is the share of useful energy in the energy taken by the device. Efficiency is at most $100\ \%$, because energy is conserved and you cannot get more useful energy out than you put in. In real devices it is often below $100\ \%$, because part of the energy taken does not end up in the defined useful purpose: for example, it becomes heat in friction or in conductors, and the heat spreads into the surroundings. When the purpose of a device is something other than heat, efficiency varies a lot: an electric motor reaches about $90\ \%$, but a lamp and a combustion engine stay clearly below $100\ \%$. In an electric heater the purpose is the heat itself: in practice all the electrical energy becomes heat that warms the room, so the efficiency is almost $100\ \%$. An ideal, lossless device would have an efficiency of $100\ \%$. The efficiencies of devices vary a lot:

Device and its useful output	typical η
incandescent lamp: light	$5\ \%$
LED lamp: light	$30\ \%$
kettle: warming of water	$90\ \%$
electric motor: motion	$90\ \%$
petrol engine: motion of a car	$25\ \%$
solar panel: electricity	$20\ \%$

The values in the table are typical rounded values, not measurements of any particular device. The real efficiency depends on the model of the device and on the way it is used. The value for the petrol engine is typical for steady motorway driving; in city driving and at other small loads the efficiency is clearly lower. The efficiency of an electric motor changes little when it runs close to the power it was designed for.

Efficiency

Efficiency η (eta) is the ratio of the useful energy to the energy taken by the device. When the device runs at constant power, the same ratio is also obtained from

powers:

$$\eta = \frac{E_\mathrm{a}}{E_\mathrm{o}} = \frac{P_\mathrm{a}}{P_\mathrm{o}}$$

where η is the efficiency (a ratio, often given as a percentage), E_a the useful energy given by the device (J), E_o the energy taken by the device (J), P_a the useful power (W) and P_o the power taken by the device (W). A ratio is converted to a percentage by multiplying it by $100\,\%$, for example $0.30 \cdot 100\,\% = 30\,\%$. The multiplication does not change the number, because $1\,\% = 0.01$ and so $100\,\% = 100 \cdot 0.01 = 1$.

The ratio of the powers is the same as the ratio of the energies, because at constant power energy is power multiplied by time (chapter *Power:* $P = \Delta E / \Delta t$). While the device runs for a time t, the energies are $E_\mathrm{a} = P_\mathrm{a} \cdot t$ and $E_\mathrm{o} = P_\mathrm{o} \cdot t$, and the time cancels:

$$\begin{aligned} \eta &= \frac{E_\mathrm{a}}{E_\mathrm{o}} \\ &= \frac{P_\mathrm{a} \cdot t}{P_\mathrm{o} \cdot t} \\ &= \frac{P_\mathrm{a}}{P_\mathrm{o}} \end{aligned}$$

So efficiency does not depend on whether energy or power is measured, or on the running time.

Conservation of energy and losses

Conservation of energy divides the energy taken by the device into useful energy and losses E_h:

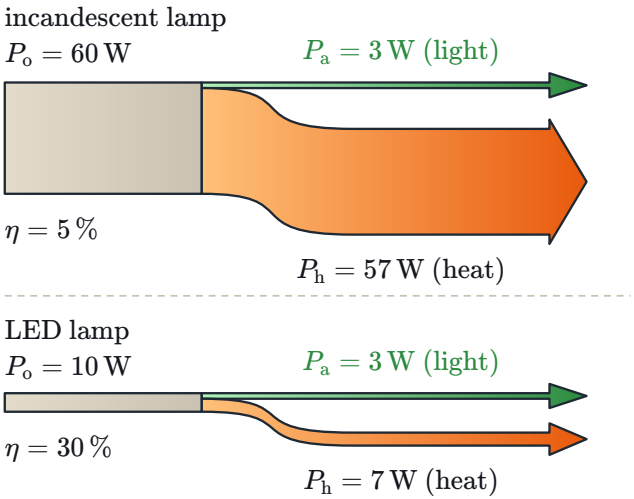
$$E_\mathrm{o} = E_\mathrm{a} + E_\mathrm{h}$$

The same holds for powers, $P_\mathrm{o} = P_\mathrm{a} + P_\mathrm{h}$. When the efficiency is known, the useful energy is $E_\mathrm{a} = \eta \cdot E_\mathrm{o}$ and the energy taken is $E_\mathrm{o} = \frac{E_\mathrm{a}}{\eta}$.

Comparing two lamps

A lamp with a low efficiency needs a higher power to get the same useful output. For a light power of $3\ \mathrm{W}$ an incandescent lamp needs $60\ \mathrm{W}$ but an LED lamp only $10\ \mathrm{W}$, because the efficiencies are $5\ \%$ and $30\ \%$. In the diagram the thicknesses of the arrows are on the same scale: the green arrows of the light power are equally thick, but the incoming arrow of the incandescent lamp is six times thicker.

In this simplified model the useful energy of a lamp is defined as the energy of visible radiation. In practice the lighting ability of a lamp is better described by its luminous flux, which takes into account that the eye is not equally sensitive to all wavelengths.



In a diagram drawn to scale you can see the efficiency directly: it is the thickness of the useful arrow as a share of the thickness of the incoming arrow. If the incoming arrow is $5.0\ \mathrm{cm}$ thick and the useful arrow $4.0\ \mathrm{cm}$, the efficiency is $4.0 : 5.0 = 0.80$, that is, $80\ \%$.

Common misconception

Many people think that energy disappears in the losses. However, energy is conserved in every device. The energy leaving an incandescent lamp is equal to the energy entering it: $3\ \mathrm{W}$ of light and $57\ \mathrm{W}$ of heat make $60\ \mathrm{W}$ in total. A loss only means that part of the energy ended up in a form that is of no use.

Another mistake is to think that a high-power device is efficient. Efficiency is a ratio, not an amount. A $100\ \mathrm{W}$ incandescent lamp gives $5\ \mathrm{W}$ of light and a $60\ \mathrm{W}$ lamp $3\ \mathrm{W}$, but both have an efficiency of $5\ \%$. A high-power device uses more energy, but power does not tell what share of the energy is useful.

Summary

- Efficiency is the share of useful energy in the energy taken by the device: $\eta = \frac{E_\mathrm{a}}{E_\mathrm{o}} = \frac{P_\mathrm{a}}{P_\mathrm{o}}$. It is often given as a percentage and is at most $100\,\%$. In real devices it is often below $100\,\%$, because part of the energy does not end up as the useful output; for an electric heater it is almost $100\,\%$.
- Energy is conserved: $E_\mathrm{o} = E_\mathrm{a} + E_\mathrm{h}$. Losses are usually heat that spreads into the surroundings.
- In an energy flow diagram the thickness of an arrow is proportional to energy or power, and efficiency is the share of the useful arrow in the incoming arrow.
- The efficiency of a device that runs steadily does not depend on running time, because the time cancels from the ratio of the energies. The way the device is used can change it: for example, the efficiency of a petrol engine is clearly lower at a small load.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Losses of a kettle

A kettle has an efficiency of $90\,\%$ and takes a power of $2.0\,\mathrm{kW}$ from the electricity grid. The kettle is on for $3.0\,\mathrm{min}$. How much electrical energy becomes losses, that is, heat that does not warm the water?

STEP 1/7

Known and unknown. Efficiency $\eta = 90\,\%$, power taken $P_\mathrm{o} = 2.0\,\mathrm{kW}$ and running time $t = 3.0\,\mathrm{min}$. We want the loss energy E_h.

STEP 2/7

Principle. The kettle runs at constant power, so the energy taken is power multiplied by time. Efficiency tells what share of the energy taken warms the water. Energy is conserved, so the rest is losses.

STEP 3/7

Rearranged formula. The energy taken, the useful energy and the losses are found in order:

$$E_{\mathrm{o}} = P_{\mathrm{o}} \cdot t$$
$$E_{\mathrm{a}} = \eta \cdot E_{\mathrm{o}}$$
$$E_{\mathrm{h}} = E_{\mathrm{o}} - E_{\mathrm{a}}$$

STEP 4/7

Unit conversions. Power to watts:

$$\begin{aligned} 2.0\,\mathrm{kW} &= 2.0 \cdot 1000\,\mathrm{W} \\ &= 2000\,\mathrm{W} \end{aligned}$$

Time to seconds:

$$\begin{aligned} 3.0\,\mathrm{min} &= 3.0 \cdot 60\,\mathrm{s} \\ &= 180\,\mathrm{s} \end{aligned}$$

Efficiency to a number: $90\,\% = 90 \cdot 0.01 = 0.90$.

STEP 5/7

Substitution.

$$\begin{aligned} E_{\mathrm{o}} &= P_{\mathrm{o}} \cdot t \\ &= 2000\,\mathrm{W} \cdot 180\,\mathrm{s} \\ &= 360\,000\,\mathrm{J} \end{aligned}$$

$$\begin{aligned} E_{\mathrm{a}} &= \eta \cdot E_{\mathrm{o}} \\ &= 0.90 \cdot 360\,000\,\mathrm{J} \\ &= 324\,000\,\mathrm{J} \end{aligned}$$

$$\begin{aligned} E_{\mathrm{h}} &= E_{\mathrm{o}} - E_{\mathrm{a}} \\ &= 360\,000\,\mathrm{J} - 324\,000\,\mathrm{J} \\ &= 36\,000\,\mathrm{J} \end{aligned}$$

STEP 6/7

Result. The energy that becomes losses is

$$\begin{aligned} 36\,000\,\mathrm{J} &= 36 \cdot 1000\,\mathrm{J} \\ &= 36\,\mathrm{kJ} \end{aligned}$$

The result is given with two significant figures, because the initial values $90\,\%$, $2.0\,\mathrm{kW}$ and $3.0\,\mathrm{min}$ are given to two figures.

STEP 7/7

Reasonableness. The losses must be $100\,\% - 90\,\% = 10\,\%$ of the energy taken, and the energy taken is

$$\begin{aligned} 360\,000\,\mathrm{J} &= 360 \cdot 1000\,\mathrm{J} \\ &= 360\,\mathrm{kJ} \end{aligned}$$

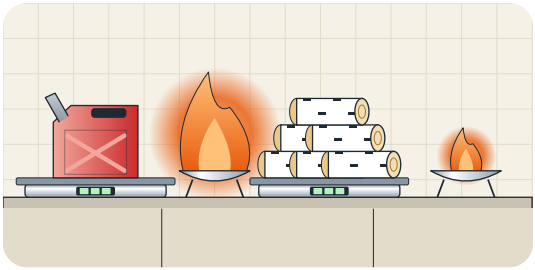
so $0.10 \cdot 360\,\mathrm{kJ} = 36\,\mathrm{kJ}$. In addition, the useful energy and the losses add up to the energy taken: $324\,\mathrm{kJ} + 36\,\mathrm{kJ} = 360\,\mathrm{kJ}$.

FY2 · CHAPTER 5

Fuels and heat of combustion $E = Hm$

Why do a kilogram of petrol and a kilogram of air-dried firewood release different amounts of energy when they burn, even though both are just burning matter?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy2-5.html`



IN THIS LESSON, YOU WILL LEARN TO

- calculate the energy released from a fuel using its heat of combustion and mass.
- compare fuels and food by energy content and explain how food energy differs from the heat of combustion.
- tell released energy apart from useful energy using efficiency.

PRIOR KNOWLEDGE

- Energy is neither created nor destroyed but transformed from one form to another
  chapter *Forms of energy and conservation of energy*
- The unit of energy is the joule, and $1\,\mathrm{kWh} = 3.6\,\mathrm{MJ}$
  chapter *Energy transformations and the joule*
- Efficiency tells you what share of the energy taken in is useful
  chapter *Efficiency*
- Power is energy per unit of time
  chapter *Power P = ΔE/Δt*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 5.

Think about it

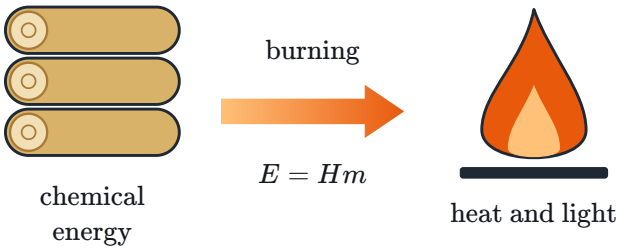
At a petrol station a car is filled with petrol, and in a cottage fireplace firewood is burnt. On the side of a chocolate bar it says "energy $2200\,\mathrm{kJ}$ per hundred grams". In all three, energy is stored in the substance, and it is released when the substance burns or when the body uses it.

> **Is the same amount of energy released from a kilogram of different substances, and how can you calculate the energy released?**

Theory

Key idea

In this chapter a fuel means a chemical fuel: a substance whose chemical energy is released by burning it. The uranium in a nuclear power plant is also called a fuel, but nuclear fuel does not burn: its energy is nuclear energy (chapter *Forms of energy and conservation of energy*). Chemical energy is energy stored in the structure of the atoms and molecules of a substance. When a fuel burns, the structure changes and energy is released as heat and light. Energy does not appear out of nothing: chemical energy is transformed into another form. The same is true of food: your body slowly "burns" food and releases energy from it.



Different fuels contain different amounts of chemical energy for their mass. The quantity that tells you this is the heat of combustion H. It is the energy released when one kilogram of fuel burns, and its unit is, for example, $\mathrm{MJ/kg}$. The heat of combustion of petrol is about $43\ \mathrm{MJ/kg}$, so about $43\ \mathrm{MJ}$ is released from one kilogram (about $43 \cdot 10^6\ \mathrm{J}$). The heat of combustion of air-dried firewood (moisture $20\text{–}25\ \%$) is only about $15\ \mathrm{MJ/kg}$, so to get the same energy you need almost three times the mass of wood.



When wood burns, its chemical energy is released as heat and light.

Photo: Tom Fisk, Pexels License

Because the heat of combustion is energy per unit of mass, the energy released is directly proportional to the mass of fuel burnt: twice as much fuel releases twice as much energy. This gives the law used to calculate the energy released.



A bucket-wheel excavator in the Garzweiler open-pit mine in Germany. The mine produces lignite, a moist coal with a low heating value.

Photo: © Raimond Spekking, CC BY-SA 4.0

Heat of combustion and released energy

The energy released when a fuel burns is

$$E = Hm$$

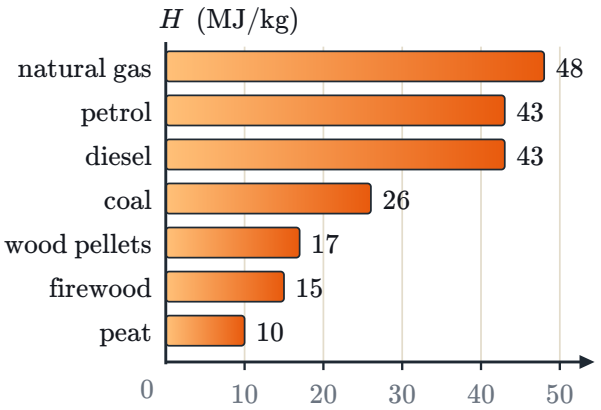
where E is the energy released (J or MJ), H is the heat of combustion of the fuel ($\mathrm{MJ/kg}$) and m is the mass of fuel burnt (kg). The mass must be in kilograms, because the heat of combustion is energy per kilogram. For example, $3.0\ \mathrm{kg}$ of

petrol releases $43\ \frac{\mathrm{MJ}}{\mathrm{kg}} \cdot 3.0\ \mathrm{kg} = 129\ \mathrm{MJ} \approx 130\ \mathrm{MJ}$, because the mass is known to two figures.

The table gives the heats of combustion of common fuels. They are net heats of combustion, meaning that the heat released when water vapour in the flue gas condenses is not counted, and energy statistics use them. The values are rounded: the real value varies from batch to batch of fuel.

Fuel	H (MJ/kg)
natural gas	48
petrol	43
diesel	43
coal	26
wood pellets	17
firewood (air-dried)	15
peat (milled peat)	10

Sources: firewood (13–$15\ \mathrm{GJ/t}$, moisture 20–$25\ \%$) and milled peat (9–$11\ \mathrm{GJ/t}$, moisture 40–$50\ \%$): Statistics Finland, *Polttoaineluokitus, määritelmät* (2017). Natural gas $48.0\ \mathrm{MJ/kg}$: IPCC, *2006 Guidelines for National Greenhouse Gas Inventories*, volume 2, table of default heating values (2006). Typical values: coal (IPCC 2006: $25.8\ \mathrm{MJ/kg}$), diesel (IPCC 2006: $43.0\ \mathrm{MJ/kg}$), petrol ($43\ \mathrm{MJ/kg}$; the IPCC 2006 value for pure petrol is $44.3\ \mathrm{MJ/kg}$, but Finnish petrol contains ethanol, which has a lower heat of combustion) and wood pellets (published typical value). The values are rounded.



Moisture and fuel quality

Wet wood gives less energy per kilogram than dry wood, because part of its mass is water, which does not burn, and in addition part of the released energy is used to vaporise this water. This is why the firewood in the table is air-dried (moisture $20–25\ \%$) and the peat is milled peat with a moisture of $40–50\ \%$. Completely dry wood would have a heat of combustion of about $19\ \mathrm{MJ/kg}$.

The energy content of food and the heat of combustion

A nutrition label gives energy per hundred grams, for example $2200\ \mathrm{kJ}$ per $100\ \mathrm{g}$. When you convert the energy to megajoules and the mass to kilograms, you get the energy content of the food per unit of mass, here $e = 22\ \mathrm{MJ/kg}$. Both the energy content of food and the heat of combustion of a fuel can be given in the unit $\mathrm{MJ/kg}$. The nutrition label describes the energy the body can use, whereas the heat of combustion of a fuel describes the energy released in burning. So they are not the same quantity: your body does not get all the energy from food that would be released if it burned, because, for example, part of the food is not absorbed from the intestine. The calculation is still the same as with the heat of combustion: the energy of food is $E = em$.

Nutritional value
per 100 g
energy 2200 kJ
example

$$2200\,\text{kJ} = 2.2\,\text{MJ}$$

$$100\,\text{g} = 0.100\,\text{kg}$$

$$e = 22\,\frac{\text{MJ}}{\text{kg}}$$

Food	kJ / 100 g	e (MJ/kg)
potato, boiled	300	3
rye bread	1000	10
milk chocolate	2200	22
butter	3000	30

Typical nutritional values (reference: Fineli, THL), rounded and varying between products.

Common misconception

- **Heat of combustion is not temperature.** Many people think the heat of combustion tells you how hot the fuel burns. It is actually energy per unit of mass.
- **Energy is not created in burning.** The chemical energy of the fuel is transformed into heat and light, but energy does not appear out of nothing.
- **Released energy is not useful energy.** In a boiler, a fireplace or an engine, part of the energy is wasted, and the useful share is calculated with efficiency (chapter *Efficiency*).

Burning carbon-containing fuels, such as wood, peat and fossil fuels, also produces carbon dioxide, which is covered in the chapter *Energy production, the environment and climate change*.

Summary

- The chemical energy of a fuel is released as heat and light when it burns. Energy is not created but transformed.

- The heat of combustion H is the energy released per kilogram of fuel, with the unit $\mathrm{MJ/kg}$.
- The energy released is $E = Hm$ when the mass m is in kilograms.
- The heat of combustion of petrol and diesel is about $43\ \mathrm{MJ/kg}$, of coal 26, of air-dried wood 15 and of peat $10\ \mathrm{MJ/kg}$.
- The kJ per $100\ \mathrm{g}$ on a food package can be converted to $\mathrm{MJ/kg}$ like a heat of combustion, but it describes the energy the body can use, not the energy released in burning.
- Released energy is not the same as useful energy: useful energy is found with efficiency.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Energy from firewood

In a fireplace, $6.0\ \mathrm{kg}$ of air-dried firewood with a heat of combustion of $15\ \frac{\mathrm{MJ}}{\mathrm{kg}}$ is burnt. How much energy is released from the wood? Give the answer in megajoules and in kilowatt-hours.

STEP 1/7

Given and find. The mass of the firewood is $m = 6.0\ \mathrm{kg}$ and from the table the heat of combustion is $H = 15\ \frac{\mathrm{MJ}}{\mathrm{kg}}$. We want the energy released, E.

STEP 2/7

Principle. The wood burns, so its chemical energy is transformed into heat. The heat of combustion tells you the energy per kilogram, so the energy released is found by multiplying the heat of combustion by the mass.

STEP 3/7

Rearranged formula. The quantity we want, E, is already alone on the left side of the equation:

$$E = Hm$$

STEP 4/7

Unit conversions. The mass is already in kilograms, so you can calculate the megajoules directly. For kilowatt-hours we use $1\,\mathrm{kWh} = 3.6\,\mathrm{MJ}$, from which $1\,\mathrm{MJ} = 1\,\mathrm{kWh} : 3.6$.

STEP 5/7

Substitution.

$$\begin{aligned} E &= Hm \\ &= 15\ \tfrac{\mathrm{MJ}}{\mathrm{kg}} \cdot 6.0\,\mathrm{kg} \\ &= 90\,\mathrm{MJ} \end{aligned}$$

In kilowatt-hours

$$\begin{aligned} E &= 90\,\mathrm{MJ} \\ &= 90 \cdot (1\,\mathrm{kWh} : 3.6) \\ &= 25\,\mathrm{kWh} \end{aligned}$$

STEP 6/7

Result. The wood releases $90\,\mathrm{MJ}$ of energy, which is $25\,\mathrm{kWh}$. The answer is given to two significant figures, because both the mass $6.0\,\mathrm{kg}$ and the heat of combustion $15\ \tfrac{\mathrm{MJ}}{\mathrm{kg}}$ are known to two figures.

STEP 7/7

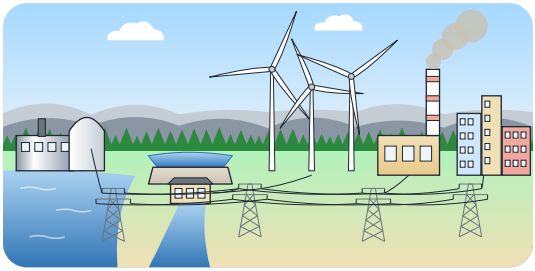
Reasonableness. One kilogram of wood gives $15\,\mathrm{MJ}$, which is about $4\,\mathrm{kWh}$, so six kilograms give about six times as much: $6 \cdot 4\,\mathrm{kWh} = 24\,\mathrm{kWh}$, which matches the result. The energy equals that of a one-kilowatt ($1\,\mathrm{kW}$) heater switched on for $25\,\mathrm{h}$. However, you do not get all of it as heat for the room, because some of the heat leaves with the flue gases (chapter *Efficiency*).

FY2 · CHAPTER 6

Ways of producing energy and how they compare

Why do nuclear power, hydropower, wind power and fuel-fired power plants all run in the Finnish electricity grid at the same time, even though one type of plant could in principle produce all the electricity?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy2-6.html`



IN THIS LESSON, YOU WILL LEARN TO

- calculate the annual energy or average power of a power plant with the formula $E = P \cdot t$.
- classify power plants by weather dependence, controllability and steady base-load use, and justify the classification.
- compare ways of producing energy using orders of magnitude and criteria.

PRIOR KNOWLEDGE

- Power tells you how quickly energy is transferred, $P = E : t$
  chapter *Power:* $P = \Delta E / \Delta t$
- The energy of a fuel is calculated with the heat of combustion, $E = Hm$
  chapter *Fuels and heat of combustion* $E = Hm$
- Efficiency tells you what fraction of the energy is useful
  chapter *Efficiency*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 4.

Think about it

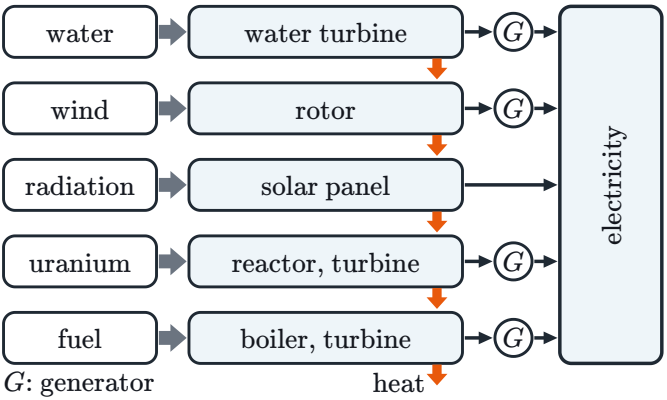
On a frosty morning at eight o'clock Finland wakes up: heating, industry and trains use more electricity than almost ever. There is little wind, and the Sun will not rise for another hour. Still, the lights are on. The electricity comes from nuclear power plants, hydropower plants, fuel-fired power plants and a little from wind.

> **Why does the electricity grid need many different kinds of power plants, instead of just one best one?**

Theory

Key idea

A power plant does not produce energy out of nothing. It converts the energy of a natural resource into electricity and often also into heat. The kinetic energy of falling water and wind, the radiant energy of the Sun, the nuclear energy of uranium and the chemical energy of fuel all travel through a turbine and generator or a solar panel into the electricity grid. Energy is conserved, but some of it always ends up as heat in the surroundings (chapter *Efficiency*).



Renewable and non-renewable energy

Water, wind and solar radiation are renewable: they do not run out when used. Biofuels such as wood are renewable when a new forest grows to replace the old one. Coal, oil and natural gas are fossil fuels, and uranium is also a non-renewable resource. Peat grows in bogs so slowly, over thousands of years, that it is treated like the fossil fuels when emissions are counted; statistics often list it as a group of its own. The emissions and climate effects of power plants are covered in chapter *Energy production, the environment and climate change*.



Turbines of the Summa wind farm in Hamina, Finland. A wind turbine turns the kinetic energy of the wind into electricity.

Photo: IA (Wikimedia Commons), CC0 1.0

Three properties: how a power plant works in the grid

Large amounts of electricity cannot be stored directly in the grid, so production must match consumption at every moment. This is why power plants are described by how their output can be influenced. The groups are useful properties, not categories that exclude each other: hydropower is highly controllable but also depends on rainfall.

Property	Typical power plants	Power
Weather-dependent	wind, solar	varies with the weather
Controllable	water, fuel	can be adjusted
Base load	nuclear power	almost constant

In a hydropower plant the flow is increased or reduced quickly, but you cannot give orders to the wind or the Sun. A nuclear power plant usually runs steadily at almost full power, but its output can also be reduced when needed: for example, the Olkiluoto units have reduced their output when the price of electricity has been very low.

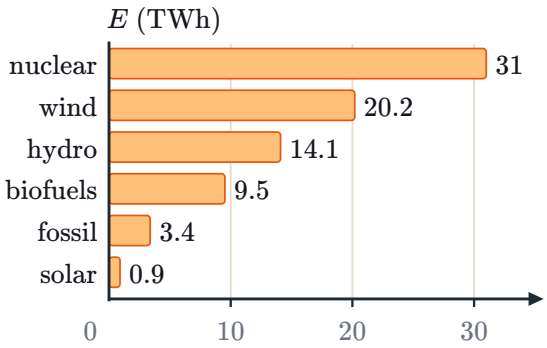
Criteria for comparison

Different ways of producing energy are compared with the same questions:

- **Cost:** how much it costs to produce the electricity.
- how large a single power plant is.
- **Transportability:** whether the energy source can be brought to the plant, like fuel, or whether the plant has to be built where the wind or flowing water is.
- **Storability:** fuel and a water reservoir can be stored, but wind and solar radiation cannot.
- **Safety:** the effect and risk of an accident.
- **Environment:** emissions, land use, nature and waste.

Electricity production in Finland

In 2024 Finland produced about $80\ \mathrm{TWh}$ of electricity in total (Statistics Finland). Nuclear power produced the most, wind power about a quarter, and the share of fossil fuels was only a few per cent. The bars are drawn to scale.





The Olkiluoto nuclear power plant in Eurajoki, Finland, in 2015. The third unit, on the left, began regular electricity production in 2023.

Photo: Hannu Huovila / TVO, CC BY 3.0

Energy of a power plant

Power tells you how quickly a power plant produces energy. When the power is constant, the energy produced is power multiplied by time:

$$E = P \cdot t$$

where E is the energy produced, P is the power of the plant and t is time. When the power is in kilowatts and the time in hours, the energy is in kilowatt-hours. An ordinary year of 365 days has $t = 8760\,\mathrm{h}$. A leap year has $8784\,\mathrm{h}$. The worked calculations of this chapter use the length of an ordinary year unless stated otherwise. Large amounts of energy are often given in gigawatt-hours and terawatt-hours: $1\,\mathrm{GWh} = 10^6\,\mathrm{kWh}$ and $1\,\mathrm{TWh} = 10^9\,\mathrm{kWh}$.

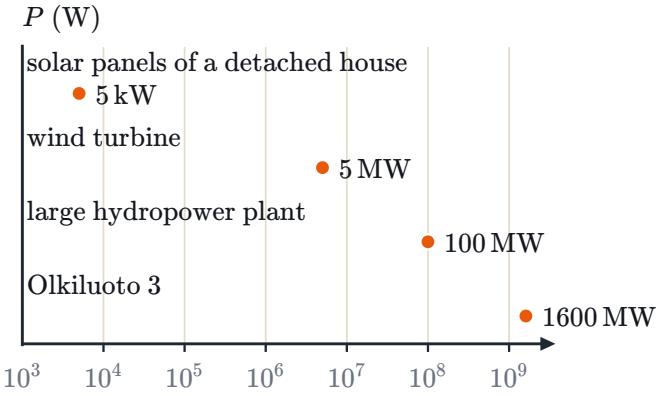
Rated power and average power

The rated power of a power plant is the highest power at which it can operate. A wind turbine does not produce at rated power all year, because the wind varies. The average power is the energy produced during the year divided by the number of hours in the year. The average power divided by the rated power tells you what fraction of the rated power is used on average.

Orders of magnitude

The power of one power plant can be a thousand times greater than that of another. This is why orders of magnitude are compared using powers of ten. On

the horizontal axis of the figure, each step from one vertical line to the next is a factor of ten, so the distance between the points tells you how many factors of ten the power increases by.



Common misconception

Rated power does not tell you how much electricity a power plant actually produces. A wind turbine with a rated power of $5\ \mathrm{MW}$ produces in a year only about a third of what it would produce if the wind always blew at the right speed. Renewable also does not mean available all the time: wind and sun are renewable but weather-dependent.

Summary

- A power plant converts the energy of a natural resource into electricity; energy is conserved, but some of it becomes waste heat.
- Power plants are described by properties that do not exclude each other: weather-dependent (wind, solar), highly controllable (water, fuel) and base load that usually runs steadily (nuclear power).
- The energy produced is $E = P \cdot t$; the average power is the annual energy divided by the number of hours in the year, $8760\ \mathrm{h}$.
- Power plants are compared by cost, power, transportability, storability, safety and environmental effects.
- Orders of magnitude are compared with powers of ten: the rated power of Olkiluoto 3 is more than three hundred times the rated power of a wind turbine.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Average power of a wind turbine

A wind turbine has a rated power of $5.0\,\mathrm{MW}$ and produces $14\,\mathrm{GWh}$ of electricity during a year. What is its average power, and what fraction of the rated power is it?

STEP 1/7

Given and find. Energy $E = 14\,\mathrm{GWh}$, time $t = 8760\,\mathrm{h}$ and rated power $P_{\mathrm{nom}} = 5.0\,\mathrm{MW}$. We want the average power P_{avg} and its share of the rated power.

STEP 2/7

Principle. Energy is power multiplied by time, $E = Pt$. The power of a wind turbine varies, so the annual energy gives the average power at which the turbine would have produced the same energy steadily.

STEP 3/7

Rearranged formula.

$$E = P_{\mathrm{avg}}\, t$$

$$P_{\mathrm{avg}} = \frac{E}{t}$$

STEP 4/7

Unit conversions.

$$14\,\mathrm{GWh} = 14 \cdot 10^{6}\,\mathrm{kWh}$$

$$\begin{aligned} 5.0\,\mathrm{MW} &= 5.0 \cdot 1000\,\mathrm{kW} \\ &= 5000\,\mathrm{kW} \end{aligned}$$

STEP 5/7

Substitution.

$$\begin{aligned} P_{\text{avg}} &= E : t \\ &= 14 \cdot 10^{6}\,\text{kWh} : 8760\,\text{h} \\ &= 1598\,\text{kW} \\ \text{share} &= P_{\text{avg}} : P_{\text{nom}} \\ &= 1598\,\text{kW} : 5000\,\text{kW} \\ &= 0.32 \end{aligned}$$

STEP 6/7

Result. The average power is

$$\begin{aligned} 1598\,\text{kW} &= 1.598\,\text{MW} \\ &\approx 1.6\,\text{MW} \end{aligned}$$

which is $32\,\%$ of the rated power, two significant figures like the energy $14\,\text{GWh}$.

STEP 7/7

Reasonableness. If the turbine ran at rated power all year, it would produce

$$\begin{aligned} &5000\,\text{kW} \cdot 8760\,\text{h} \\ &\quad = 43\,800\,000\,\text{kWh} \\ &\quad = 43.8 \cdot 10^{6}\,\text{kWh} \\ &\quad = 43.8\,\text{GWh} \end{aligned}$$

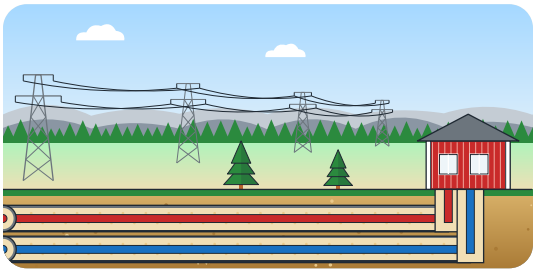
and $14 : 43.8 = 0.32$. The share is less than one, because the wind does not always blow.

FY2 · CHAPTER 7

Transferring energy: electricity and heat

Why is the voltage in power lines hundreds of kilovolts when a wall socket gives only 230 V, and why are district heating pipes wrapped in thick insulation?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy2-7.html`



IN THIS LESSON, YOU WILL LEARN TO

- explain how electrical energy and thermal energy are transferred from the power plant to the user.
- calculate the energy that arrives and the energy that is lost from a loss percentage.
- explain why long-distance electricity transmission uses high voltage.

PRIOR KNOWLEDGE

- Power tells the rate at which energy is transferred: $P = \Delta E : \Delta t$
  chapter *Power:* $P = \Delta E / \Delta t$
- Efficiency is the share of the energy used that is obtained as useful energy
  chapter *Efficiency*
- Energy does not disappear but changes form, often into heat
  chapter *Forms of energy and conservation of energy*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 4.

Think about it

The wall socket at home gives $230\ \mathrm{V}$, but in the wires of the power lines across the field the voltage is hundreds of kilovolts. In winter steam rises from a district heating manhole at the edge of the street: hot water runs underground for kilometres.

> **Why is energy transferred like this, and how much of it is wasted on the way?**

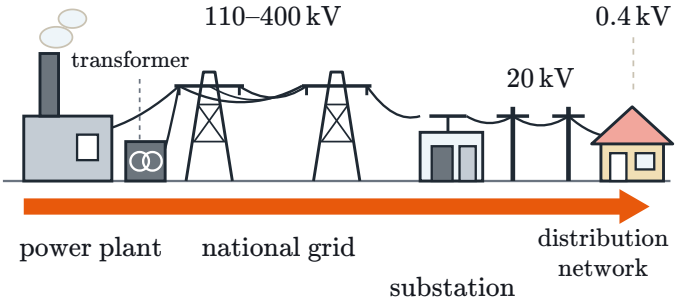
Theory

Key idea

Energy is often produced far from where it is used: the power plant is outside the city, but the heater and the wall socket are at home. So the energy has to be transferred to the user. Energy can be transferred from place to place in many ways. This chapter looks at two ways that are important for society's energy networks: transferring in the electricity grid and transferring **thermal energy** in the water pipes of the district heating network.

The electricity grid

The **electricity grid** carries electrical energy from the power plant to the home. Long distances are covered by the power lines of the national grid, whose voltage is $110–400\ \mathrm{kV}$. Closer to the user the voltage is first lowered to $20\ \mathrm{kV}$ and finally to the $0.4\ \mathrm{kV}$ of the low-voltage network, from which the sockets in homes get $230\ \mathrm{V}$. The voltage is changed in transformers.



Why long distances use high voltage

The same power can be transferred in two ways: with a low voltage and a large electric current, or with a high voltage and a small electric current. A wire gets hotter the larger the current in it is, and the energy used for heating is a loss. So long distances are covered with a high voltage, which keeps the current and the loss small. The exact connection between voltage, current and loss belongs to later courses on electricity.

A transformer raises the voltage at the power plant and lowers it again close to the user. In a good transformer the loss is very small.

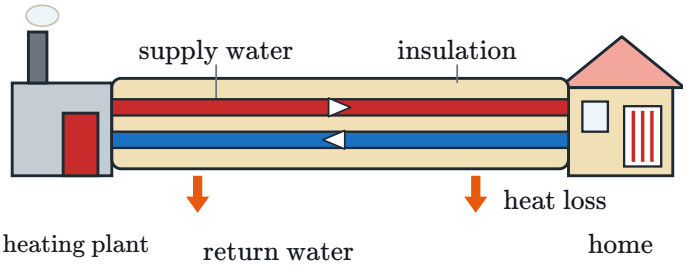


A substation transformer in Muurame, Finland. At a substation the transformer lowers the high voltage of the power line to the voltage of the distribution network.

Photo: Antti Leppänen, CC BY 4.0

District heating

transfers energy as hot water. The water heated at the heating plant flows in the supply pipe to buildings and gives heat to the building's heating system. The cooled water returns in the return pipe to be heated again. The pipes are underground in thick insulation so that as little heat as possible goes into the surrounding ground.



Transmission loss

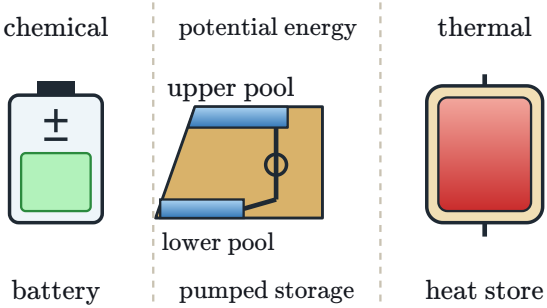
In the electricity grid some of the electrical energy is converted into internal energy in the wires and other equipment, and this is transferred as heat to the

surroundings. In the district heating network, energy is transferred as heat from the hot water through the pipes and insulation to the surroundings. In both cases the users get less energy than is fed into the network. This is called . The energy does not disappear (chapter *Forms of energy and conservation of energy*), but the loss is useless to the user.

In Finland the losses of the electricity grid in 2024 were about $4\ \%$ of electricity consumption, when consumption was about $83\ \mathrm{TWh}$ (Finnish Energy, Sähkövuosi 2024). In this chapter consumption is used as the energy sent into the electricity grid, so $4\ \%$ is a share of it. About $36\ \mathrm{TWh}$ was supplied to the district heating network in 2024 and about $32.7\ \mathrm{TWh}$ was used by customers, so about $9\ \%$ was left on the way (Finnish Energy, Kaukolämpötilasto 2024; the share is calculated from the difference, so it is an estimate). The loss of district heating is relatively larger because hot water is clearly warmer than its surroundings: the greater the temperature difference to the surroundings, the more heat escapes, and heat gets through the insulation along the whole length of the pipework.

Transferring energy in time: storage

Energy is also transferred in time. Wind power and solar electricity are produced when the wind blows or the sun shines, but electricity is needed at other times too. A store takes in energy when there is too much of it and gives it out later.



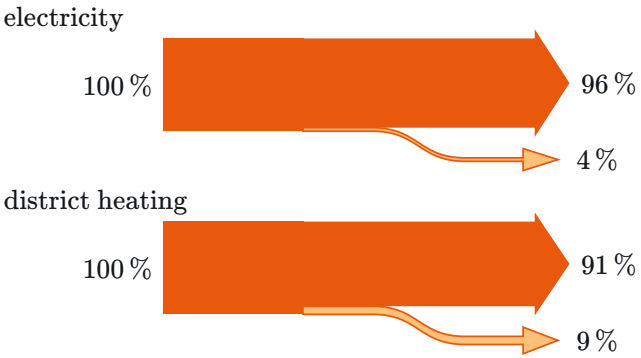
In a battery the energy is stored as chemical energy. In a pumped-storage power plant water is pumped to an upper reservoir, and the water flowing down turns the machines of the power plant. In a heat store hot water stays in an insulated tank, and it is used for example in the district heating network. Electricity is not stored in large amounts as such but by converting it into another form of energy and back, and in every conversion some of the energy turns into heat.

Transmission loss

The loss is the difference between the energy sent into the network and the energy that arrives. The efficiency of the transfer (chapter *Efficiency*) is the share of the sent energy that arrives:

$$E_{\text{loss}} = E_{\text{sent}} - E_{\text{arrived}}$$
$$\eta = \frac{E_{\text{arrived}}}{E_{\text{sent}}}$$

where E_{sent} is the energy fed into the network, E_{arrived} the energy that reached the user, E_{loss} the energy transferred as heat to the surroundings on the way and η the efficiency of the transfer. Efficiency is given as a percentage or as a decimal number.



In the figure the width of a stream shows the amount of energy. In the electricity grid $4\,\%$ of the sent energy (the consumption) is lost and $96\,\%$ arrives, in district heating $9\,\%$ and $91\,\%$.

Calculating from a loss percentage

A loss percentage is always taken of the sent energy. If the loss is $9.0\,\%$, then $100\,\% - 9.0\,\% = 91.0\,\%$ of the sent energy arrives, and $91.0\,\% = 91.0 : 100 = 0.910$. The energy you need is solved from the formula: $E_{\text{arrived}} = \eta\, E_{\text{sent}}$ and $E_{\text{sent}} = E_{\text{arrived}} : \eta$.

The same applies to power: the power that arrives is η times the power sent. Energy and power are connected by the formula $P = \frac{E}{t}$ (chapter *Power:* $P = \Delta E / \Delta t$), so with a steady loss power P_{loss} the energy lost during a time t is $E_{\text{loss}} = P_{\text{loss}} \cdot t$.

Common misconception

Many people think that energy disappears in transmission loss. Energy does not disappear. In the electricity grid some of the electrical energy is converted into internal energy in the wires, and in the district heating network energy passes from the hot water through the pipes. In both cases the energy is finally transferred as heat to the surroundings, and the user gets no benefit from it.

Another mistake is to take the loss percentage of the energy that arrives. The percentage must be taken of the sent energy: if the loss is $9.0\ \%$ and $245\ \mathrm{MWh}$ arrives, the sent energy is not $245\ \mathrm{MWh} + 0.090 \cdot 245\ \mathrm{MWh}$ but $245\ \mathrm{MWh} : 0.910$.

A third mistake is to think that the loss is subtracted from the percentage as a unit: a loss of $4\ \%$ does not mean $4\ \mathrm{MWh}$ but four hundredths of the sent energy.

Summary

- Electrical energy is transferred in the electricity grid, long distances at high voltage, because then the current is small and the loss caused by heating of the wires is small. Transformers raise and lower the voltage.
- Thermal energy is transferred in the district heating network as hot water in insulated pipes. In transmission loss, electrical energy is converted into internal energy in the wires, and energy passes from the hot water through the pipes; in both, the energy is transferred as heat to the surroundings. In Finland the losses were about $4\ \%$ in the electricity grid and about $9\ \%$ in the district heating network (Finnish Energy, 2024).
- Transmission loss $E_{\mathrm{loss}} = E_{\mathrm{sent}} - E_{\mathrm{arrived}}$ and efficiency of the transfer $\eta = E_{\mathrm{arrived}} : E_{\mathrm{sent}}$. The loss percentage is taken of the sent energy.
- Stores (battery, pumped-storage plant, heat store) transfer energy in time, but in them too some of the energy turns into heat.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: District heating losses

A district heating network transfers $245\ \mathrm{MWh}$ of thermal energy per week from the heating plant to a district of the city (an example situation). The loss of the network is $9.0\ \%$ of the sent energy. How much energy arrives and how much is lost?

STEP 1/7

Given and find. $E_{\text{sent}} = 245\ \mathrm{MWh}$ and the loss is $9.0\ \%$ of the sent energy. We want the energy that arrives, E_{arrived}, and the loss E_{loss}.

STEP 2/7

Principle. Energy does not disappear, so the sent energy equals the arriving and the lost energy together. When $9.0\ \%$ is lost, the remaining $91.0\ \%$ arrives, so the efficiency is $\eta = 0.910$.

STEP 3/7

Rearranged formula.

$$E_{\text{arrived}} = \eta\, E_{\text{sent}}$$
$$E_{\text{loss}} = E_{\text{sent}} - E_{\text{arrived}}$$

STEP 4/7

Unit conversions. The efficiency is obtained from the percentage:

$$\begin{aligned}\eta &= 100\,\% - 9.0\,\% \\ &= 91.0\,\% \\ &= 91.0 : 100 \\ &= 0.910\end{aligned}$$

The energies are in the same unit, megawatt-hours, so no other conversion is needed.

STEP 5/7

Substitution.

$$\begin{aligned} E_{\text{arrived}} &= \eta\, E_{\text{sent}} \\ &= 0.910 \cdot 245\,\text{MWh} \\ &= 222.95\,\text{MWh} \\ E_{\text{loss}} &= E_{\text{sent}} - E_{\text{arrived}} \\ &= (245 - 222.95)\,\text{MWh} \\ &= 22.05\,\text{MWh} \end{aligned}$$

STEP 6/7

Result. About $223\,\text{MWh}$ arrives and about $22\,\text{MWh}$ is lost. The arriving energy is given with three significant figures because the efficiency 0.910 and the sent energy $245\,\text{MWh}$ are known to three figures. The loss is given with two significant figures because the loss percentage $9.0\,\%$ is known only to two figures.

STEP 7/7

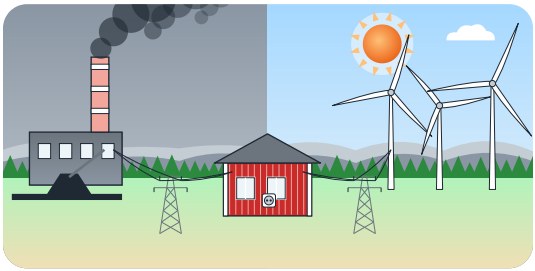
Reasonableness. Ten per cent of $245\,\text{MWh}$ would be $24.5\,\text{MWh}$, and $9.0\,\%$ is slightly less, so a loss of $22\,\text{MWh}$ is of the right size. Also $222.95\,\text{MWh} + 22.05\,\text{MWh} = 245\,\text{MWh}$, so the energy is conserved.

FY2 · CHAPTER 8

Energy production, the environment and climate change

Why does the way we produce electricity decide how much carbon dioxide ends up in the atmosphere, even though the electricity from a socket at home is always the same?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy2-8.html`



IN THIS LESSON, YOU WILL LEARN TO

- explain the greenhouse effect and why burning fossil fuels strengthens it.
- calculate an emission from the emission factor and the energy, and compare ways of producing energy.
- justify a choice using several criteria for comparison.

PRIOR KNOWLEDGE

- When a fuel burns, chemical energy is released, and energy is measured in joules and kilowatt-hours
  chapter *Fuels and heat of combustion* $E = Hm$
- Ways of producing energy differ in power, availability and environmental effects
  chapter *Ways of producing energy and how they compare*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 4.

Think about it

Before industrialisation the atmosphere contained about 280 ppm of carbon dioxide, that is, 280 molecules per million molecules of the atmosphere. In 2024 the global mean concentration was about 424 ppm (WMO Greenhouse Gas Bulletin 2025), and the rise is largely caused by burning fossil fuels for electricity, heat and transport. Even so, carbon dioxide is less than half a part per thousand of the atmosphere.

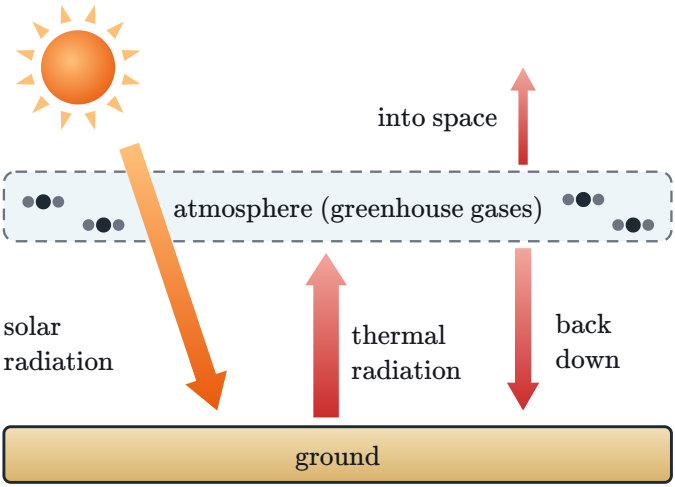
> **How can such a small part of the atmosphere change the temperature of the whole Earth, and why does it matter how electricity is produced?**

Theory

Key idea

The Sun warms the ground by radiating energy to it, as visible light and near-infrared radiation. The atmosphere lets most of this radiation through. The warm ground sends energy back into space as thermal radiation, and the in the atmosphere absorb it. The most important ones are water vapour, carbon dioxide $\mathrm{CO_2}$ and methane $\mathrm{CH_4}$. They radiate the energy out again in all directions, so part of it returns to the ground.

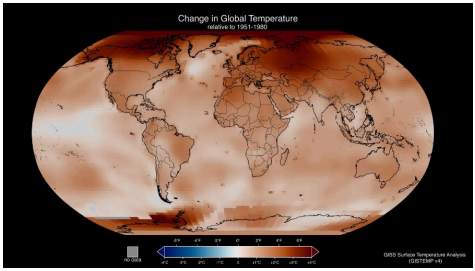
So greenhouse gases let sunlight through but slow down the escape of thermal radiation. The ground stays warmer than it would be without them. A simple radiative balance model of the Earth gives an effective radiating temperature of about $-18\,^{\circ}\mathrm{C}$, but the mean surface temperature is about $+15\,^{\circ}\mathrm{C}$. The difference shows how much the matters. The value $-18\,^{\circ}\mathrm{C}$ should not, however, be read as an exact prediction of the surface temperature if the greenhouse gases were removed, because clouds and ice cover, for example, would change too. The natural greenhouse effect keeps the Earth habitable, because without it the water in the oceans would be frozen.



The problem is that the effect is getting stronger. When people add carbon dioxide and methane to the atmosphere, thermal radiation escapes more slowly and the mean temperature of the Earth rises. This long-term change is called . The concentration of carbon dioxide in the atmosphere has grown from about $280\ \mathrm{ppm}$ (before industrialisation) to about $424\ \mathrm{ppm}$ (global mean in 2024, WMO Greenhouse Gas Bulletin 2025). Even so, carbon dioxide is less than half a part per

thousand of the air. A small share is enough, because nitrogen and oxygen, the main gases of the air, hardly absorb thermal radiation: it is the few greenhouse gas molecules that decide how easily thermal radiation escapes.

The greenhouse effect is treated with calculations on the FY3 course in the chapter *Heat transfer, energy balance and climate change*. Here it is enough to understand what the gases do to radiation and where the extra gases come from.



The mean temperature of 2025 compared with the 1951–1980 average, according to NASA's GISTEMP data. Red means warmer than the reference period and blue colder. Grey ("no data" in the key) marks places with no measurements. The map is a visualisation of measured data, not a photograph.

Photo: NASA Scientific Visualization Studio / NASA GISS (GISTEMP v4), public domain (NASA)

Fuels link energy and climate

A fossil fuel, such as coal, oil or natural gas, is chemical energy from ancient living things: coal formed mostly from plants, oil and natural gas from marine plankton. When such a carbon-containing fuel burns completely, its carbon combines with oxygen: $\mathrm{C} + \mathrm{O_2} \rightarrow \mathrm{CO_2}$. The chemical energy is released as heat (chapter *Fuels and heat of combustion* $E = Hm$), and the carbon dioxide leaves with the flue gases into the atmosphere. The more energy we get by burning carbon-containing fuels, the more carbon dioxide is released. Not every fuel contains carbon: when hydrogen burns, only water is formed.

The carbon of a fossil fuel has been underground for millions of years, so burning it adds carbon dioxide to the atmosphere. The carbon of biomass, for example wood, was bound into plants only during the last few decades. Even so, burning biomass is not emission-free: the carbon dioxide from burning stays in the atmosphere until new growth binds it back, which takes decades and requires that the plants really grow back. In addition, growing, harvesting and transport cause their own emissions.



North-western Europe at night, photographed from the International Space Station in 2011. The lights of the cities can be seen from space.

Photo: NASA, Expedition 28 crew (Mike Fossum), public domain

Ways of producing energy are compared by life-cycle emissions

Emissions come from more than just burning. A power plant must be built from steel, concrete and copper, the fuel must be mined or collected and transported, and in the end the plant is dismantled. That is why ways of producing energy are compared by **life-cycle emissions**: the emissions of all stages are added together and divided by the energy produced over the whole life cycle.

For comparison, methane and other greenhouse gases are converted into carbon dioxide equivalents (CO_2-eq.), that is, the amount of carbon dioxide that has the same warming effect.

Emissions are not the only criterion

Besides the effect on the climate, ways of producing energy are weighed by land use, transport, waste and safety:

- **Coal and natural gas:** the largest emissions. The fuels are mined and transported by ship, train or pipeline, and the flue gases also contain other pollutants.
- **Biomass:** takes up forest and field area. Biomass is bulky and its energy content per unit mass is small, so it is not worth transporting far.
- **Solar and wind power:** need a lot of area per unit of energy produced, and wind power affects the landscape and birds. Production depends on the weather, so regulating power or storage is needed.
- dams and reservoirs change the river environment and block the passage of fish, but production can be regulated.
- needs little area, and only a small mass of fuel, so there is little transport. Spent fuel is radioactive waste that must be isolated for a long time, and the consequences of an accident could be serious, although the risk is small.

A produces and uses energy so that meeting today's needs does not take away the possibilities of future generations. There is no best power plant: the choice weighs the climate effect, land use, waste, safety, price and whether electricity is always available (chapter *Ways of producing energy and how they compare*).

Carbon dioxide emission

The emission is obtained by multiplying the energy used or produced by the emission factor:

$$m = e \cdot E$$

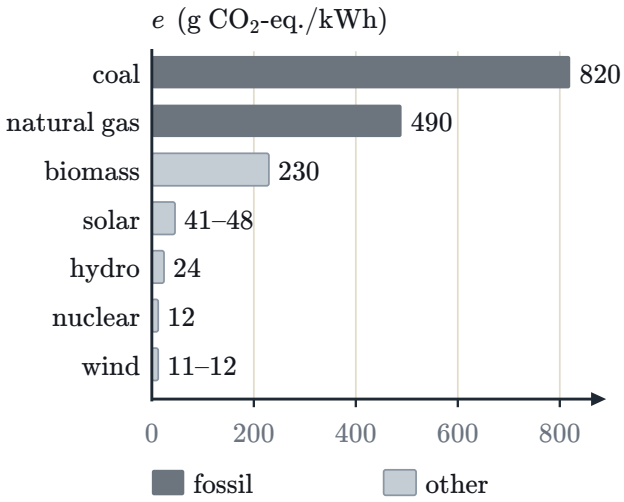
where m is the emission (g or kg of carbon dioxide equivalent), e is the **emission factor** ($\mathrm{g/kWh}$) and E is the energy (kWh).

The emission factor tells how many grams of carbon dioxide equivalent are produced per kilowatt-hour. The units carry through the calculation: when $\mathrm{g/kWh}$ is multiplied by kilowatt-hours, grams are left. The kilowatt-hour is a unit of energy, $1\,\mathrm{kWh} = 1000\,\mathrm{W} \cdot 3600\,\mathrm{s} = 3.6\,\mathrm{MJ}$. The emission of a fuel is calculated in the same way when the factor is, for example, $\mathrm{g/MJ}$ and the energy is in megajoules.

Way of producing energy	e (g/kWh)
Coal	820
Natural gas	490
Biomass	230
Solar power	41–48
Hydropower	24
Nuclear power	12
Wind power	11–12

The source of the table is the IPCC Fifth Assessment Report (AR5, 2014), Working Group III, Annex III: medians of life-cycle emissions, rounded, in the unit grams of carbon dioxide equivalent per kilowatt-hour. The lower limit for solar power is for rooftop panels and the upper limit for utility-scale plants; the lower limit for wind power is onshore and the upper limit offshore. The figure for biomass applies to biomass grown for energy. It includes the emissions of growing, harvesting, transport and the power plant, but the carbon dioxide from burning is counted as zero, because new growth binds carbon back.

This is an accounting assumption used in the emission comparison, not a claim that burning biomass releases no carbon dioxide. The real climate impact also depends on where the biomass comes from, on changes in carbon stocks and on how quickly the carbon is bound back into vegetation. So the figure does not show that the binding takes decades.



In the figure all the bars are drawn to the same scale. The dark bars are fossil fuels and the light bars are other ways of producing energy. The emission factor of coal is about 70 times that of nuclear and wind power ($820 : 12 \approx 68$). The bars of nuclear and wind power are narrow but not zero: their life cycles also include building and transport.

Common misconception

"Wind, solar and nuclear power are emission-free." They are low-emission, not emission-free: building the plants, making the fuel and transport cause emissions, about $12\ \mathrm{g/kWh}$ for wind and nuclear power. For coal the figure is about $820\ \mathrm{g/kWh}$.

"The greenhouse effect is a harmful phenomenon." The natural greenhouse effect is necessary, because without it the ground would be much colder than now: a simple radiative balance model gives the Earth an effective radiating temperature of only about $-18\ ^\circ\mathrm{C}$. The problem is the strengthening of the effect when carbon dioxide and methane are added to the atmosphere.

Summary

- Greenhouse gases let most of the Sun's radiation through but slow down the escape of thermal radiation from the ground. A simple radiative balance model gives the Earth an effective radiating temperature of about $-18\ ^\circ\mathrm{C}$, but the mean surface temperature is about $+15\ ^\circ\mathrm{C}$. The difference shows the importance of the natural greenhouse effect.
- Burning fossil fuels adds carbon dioxide to the atmosphere, strengthens the greenhouse effect and leads to climate change.

- The emission is the emission factor times the energy: $m = e \cdot E$. Ways of producing energy are compared by life-cycle emission factors, which range from about $12\ \mathrm{g/kWh}$ (nuclear and wind power) to about $820\ \mathrm{g/kWh}$ (coal).
- In the emission factor of biomass, the carbon dioxide from burning is zero as an accounting assumption; the real climate impact depends on where the biomass comes from, on changes in carbon stocks and on how quickly the carbon is bound back.
- The emission factor does not tell everything: land use, transport, waste, safety and the availability of electricity are weighed separately when looking for a sustainable energy economy.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Emissions of a family's electricity

An example family uses $4800\ \mathrm{kWh}$ of electricity a year. The electricity is produced with natural gas, whose life-cycle emission factor is $490\ \mathrm{g/kWh}$. How many tonnes of carbon dioxide equivalent are produced per year by the family's electricity use? (The family is an invented example, not statistics.)

STEP 1/7

Given and find. $E = 4800\ \mathrm{kWh}$ and $e = 490\ \mathrm{g/kWh}$. We want the emission m in tonnes.

STEP 2/7

Principle. The emission is directly proportional to the energy produced, and the emission factor gives the emission per kilowatt-hour. So the life-cycle emission is obtained by multiplying the energy by the emission factor.

STEP 3/7

Rearranged formula.

$$m = e \cdot E$$

STEP 4/7

Unit conversions. The kilowatt-hours cancel, so the result comes out in grams. Grams are converted into tonnes: $1\,\mathrm{g} = 0.001\,\mathrm{kg}$ and $1\,\mathrm{kg} = 0.001\,\mathrm{t}$.

STEP 5/7

Substitution.

$$\begin{aligned} m &= e \cdot E \\ &= 490\,\mathrm{g/kWh} \cdot 4800\,\mathrm{kWh} \\ &= 2\,352\,000\,\mathrm{g} \\ &= 2\,352\,000 \cdot 0.001\,\mathrm{kg} \\ &= 2352\,\mathrm{kg} \\ &= 2352 \cdot 0.001\,\mathrm{t} \\ &= 2.352\,\mathrm{t} \end{aligned}$$

STEP 6/7

Result. The emission is $m \approx 2.4\,\mathrm{t}$ of carbon dioxide equivalent per year, to two significant figures like the emission factor $490\,\mathrm{g/kWh}$.

STEP 7/7

Reasonableness. With rounded numbers

$$\begin{aligned} &500\,\mathrm{g/kWh} \cdot 5000\,\mathrm{kWh} \\ &\quad = 2\,500\,000\,\mathrm{g} \\ &\quad = 2\,500\,000 \cdot 10^{-6}\,\mathrm{t} \\ &\quad = 2.5\,\mathrm{t} \end{aligned}$$

which is close to the result. If the same electricity were produced with wind power, the emission would be

$$
\begin{aligned}
&12\,\mathrm{g/kWh} \cdot 4800\,\mathrm{kWh} \\
&\quad = 57\,600\,\mathrm{g} \\
&\quad = 57\,600 \cdot 0.001\,\mathrm{kg} \\
&\quad \approx 58\,\mathrm{kg}
\end{aligned}
$$

about 41 times smaller ($490 : 12 \approx 41$).

Theory and worked examples from Ruukkilabra

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